Retail Sales Analytics Dashboard

CAPSTONE PROJECT · DATA VISUALISATION & ANALYTICS

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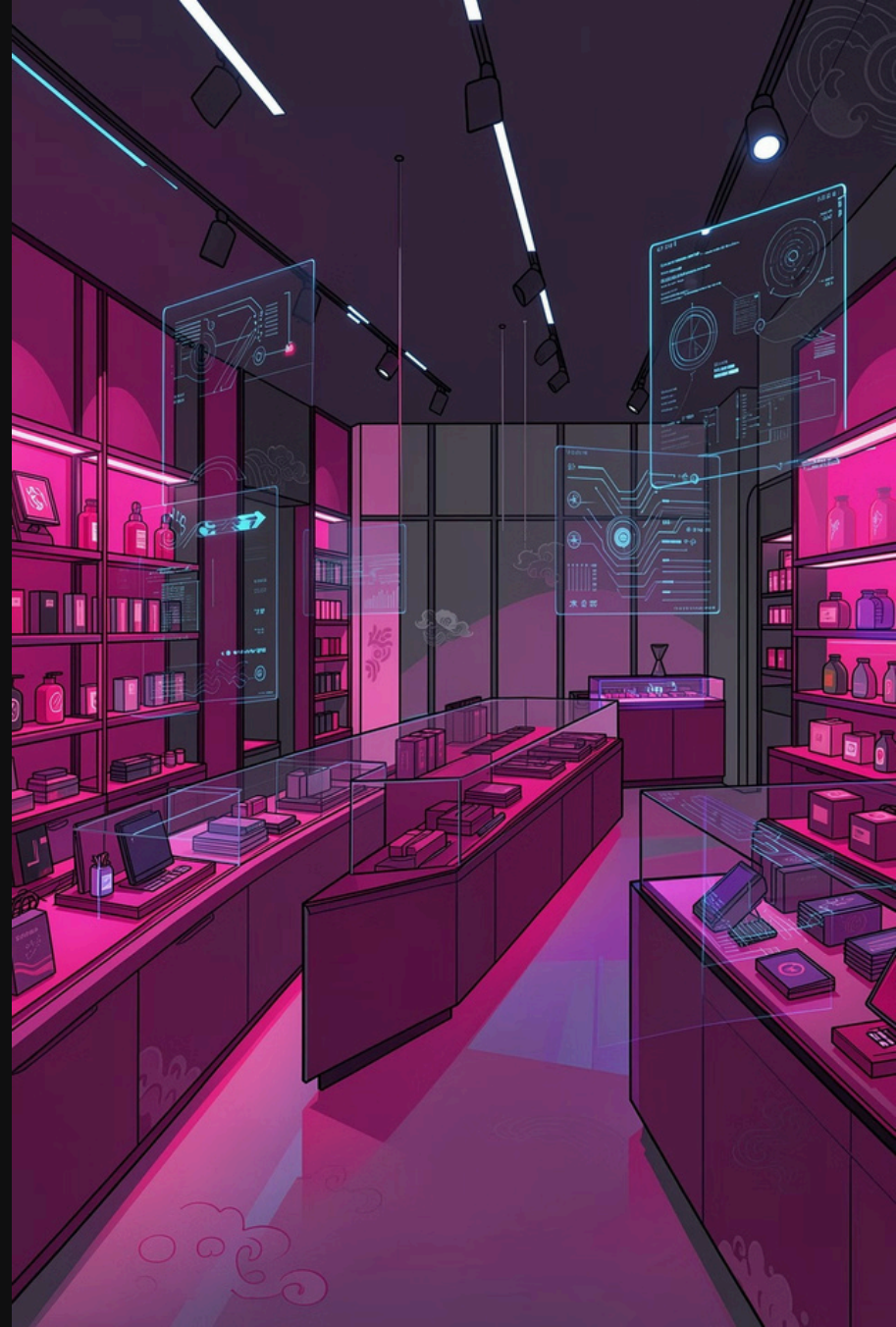
Sector: Retail

Tools: Python · Pandas · Tableau · GitHub

Dataset: 5,000+ Transactional Records



Context & Problem Statement

The Business Challenge

Retailers operate with razor-thin margins and face mounting pressure from shifting consumer behaviour, discount wars, and regional demand variability. Without structured analytics, critical profit leakages go undetected and growth opportunities remain untapped.

Core Question

What drivesprofitability across regions, categories, and time periods — and where are we losing margin?

Project Objective

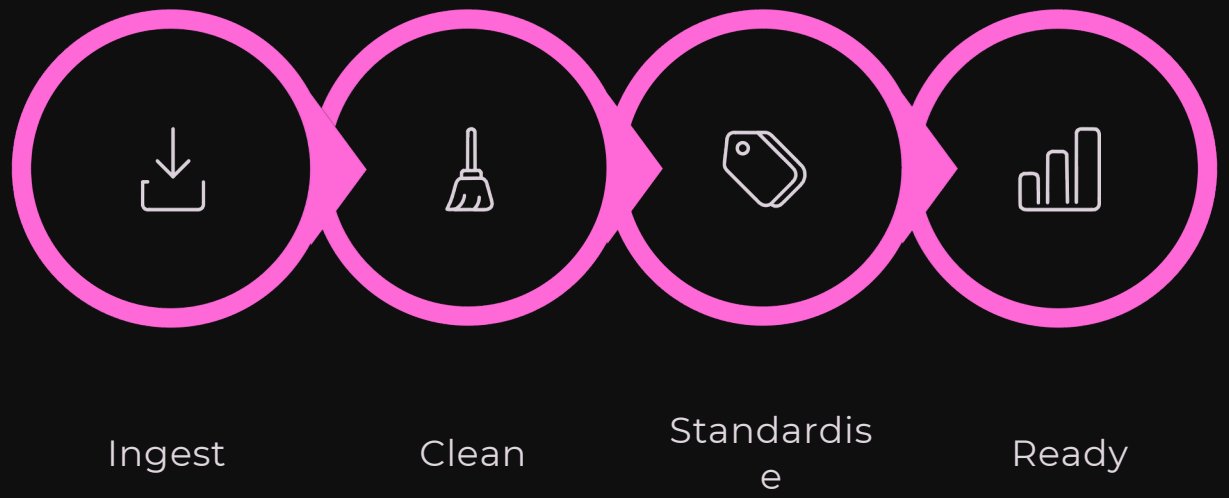
Leverage Python & Tableau to surface actionable insights from 5,000+ retail transactions and inform strategic decisions.

Data Engineering Pipeline

DATASET: RETAIL TRANSACTIONAL RECORDS

Dataset Overview

Rows: 5,000+ records
Columns: Order Date, Sales, Profit, Quantity, Region, State, Category, Product Name, Discount
Time Period: Multi-year retail transactions
Source: Structured CSV dataset (Kaggle / Internal)



All cleaning steps were executed in Python (Pandas) within Jupyter Notebook, ensuring a reproducible, audit-ready pipeline.

KPI Framework

Fourheadlinemetricsanchoreveryanalyticalview — translating raw transactions into executive-level performance signals.

Total Sales

$ 2.30 Million

Top-line revenue generated across all regions, categories, and time periods.

Total Profit

$ 286.40K

Net profitability after discounts — the true health indicator of the retail portfolio.

Total Orders

9994

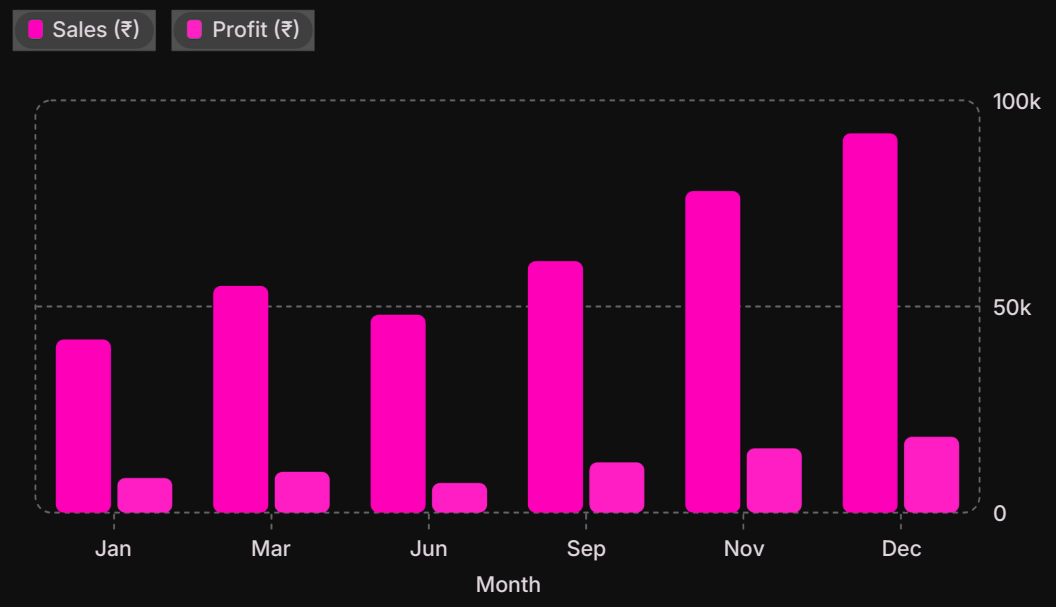
Volume of transactions processed — a proxy for customer engagement and demand velocity.

Profit Margin %

12.47%

Ratio of profit to sales — the single most critical lever for retail sustainability.

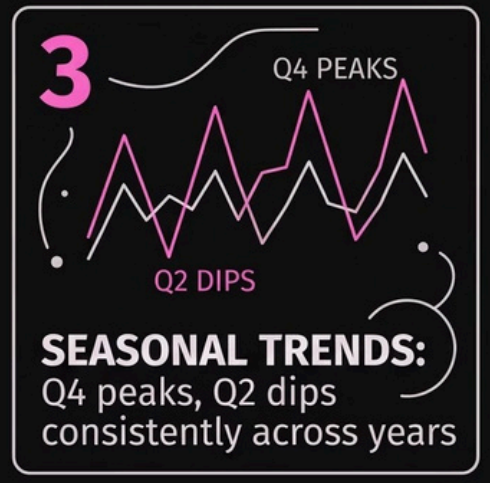
Key EDA Insights



What the Data Reveals

- **Q4 Surge:** November–December account for disproportionate revenue share — clear seasonality signal.
- **West Region** leads in sales volume; South Region shows highest profit margin efficiency.
- **Technology** category dominates revenue; Furniture carries the highest discount burden.
- **Discount vs Profit:** Discounts above 20% consistently invert profit margins across all categories.

Advanced Analysis



Analytical Methodology

Advancedtechniquesappliedin Python uncovered non-obvious patterns hidden within raw transactional data:

- Pearson correlation coefficient between Sales & Profit
- Discount threshold analysis using scatter regression
- Year-over-year seasonal decomposition
- Pareto segmentation of SKU-level profitability

ⓘ Key finding: A 1% reduction in average discount rate is projected to improve overall profit margin by ~1.8 percentage points.

Tableau Dashboard Overview

Executive View

High-level KPIs, regional heatmap, and monthly trend line — designed for C-suite consumption.

Operational View

Category drill-down, product-level profitability, and discount impact charts for category managers.

Interactive Filters

Dynamic slicersforRegion,Category, Sub-Category, and Date Range — enabling self-service exploration.

Top Strategic Insights

EXECUTIVE SUMMARY

1

West & East Regions Drive Revenue

Together, West and East account for 68% of total sales. South Region, despite lower volume, delivers the strongest margin per order — a growth priority.

2

Discount-Driven Margin Leakage

Furniture and Office Supplies sub-categories suffer persistent margin erosion from blanket discount strategies exceeding 25%.

3

Revenue Concentration Risk

Top 50 products generate over 60% of total profit — signalling over-reliance and the need for portfolio diversification.

4

Q4 Seasonal Opportunity

November–December present a concentrated demand window. Strategic inventory build and targeted promotions can amplify seasonal yield significantly.

Strategic Recommendations

Consulting-grade recommendations prioritised by impact and feasibility:

Recommendation	Rationale	Priority
Implement discount guardrails (max 20%) for Furniture & Office Supplies	Discounts beyond 20% invert margins; guardrails protect bottom-line without sacrificing volume	**Critical**
Double down on West & East Region expansion	High-volume markets with proven demand signals offer the lowest cost-to-grow	**High**
Build Q4 inventory buffer for top 50 SKUs	Prevent stockouts during peak demand; maximise seasonal revenue capture	**High**
Develop South Region as a margin excellence model	Highest profit-per-order ratio — replicate pricing discipline across regions	**Medium**
Diversify product portfolio beyond top-50 SKUs	Reduce concentration risk and unlock latent demand in long-tail categories	**Medium**

Business Impact & Next Steps

12%

Revenue Uplift

From seasonal&regionaltargeting

1.8x

Per 1% reduction in avg. discount

20%

Inventory Optimisation

Reduction in overstock exposure

Future Enhancements

01

AI-Powered Demand Forecasting

IntegrateML models (ARIMA/Prophet) for rolling sales forecasts by region and SKU.

02

Customer Segmentation Analytics

RFMmodellingto identifyhigh-value customer cohorts and personalise engagement.

03

Real-Time Dashboard Integration

Connectlive POSdata toTableaufor continuous monitoring and automated alerts.

This project demonstrates end-to-end analytical thinking — from raw data to boardroom-ready insights. Thank you for your time and evaluation.